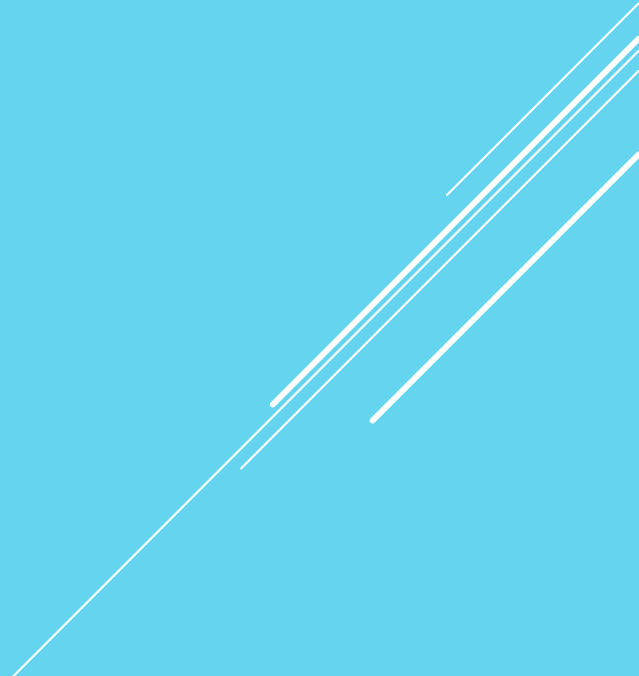
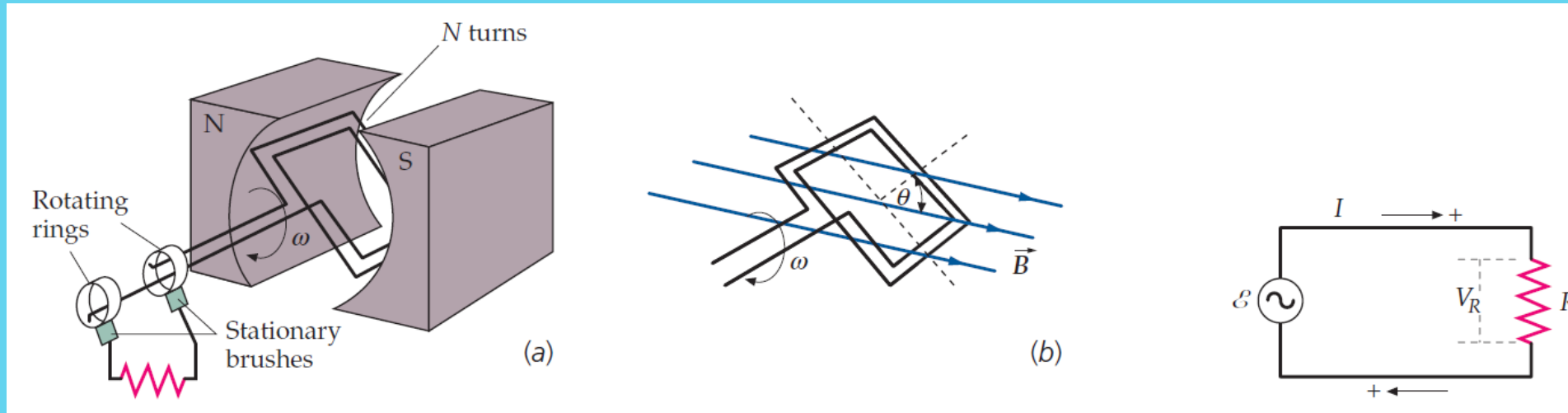


Circuitos de corriente alterna

1. Corriente alterna. Potencia consumida. Valores eficaces. Impedancia.
 2. Fasores y notación compleja.
 3. Asociación de impedancias. Resonancia.
- 
- Several white lines of varying lengths and slopes are positioned in the bottom right corner of the slide, creating a modern, abstract graphic element.

Corriente alterna con una resistencia



La figura a) muestra un generador de corriente alterna (ac). La fem de este generador está dada por

$$\mathcal{E} = \mathcal{E}_{\text{peak}} \cos \omega t$$

Donde ω es la velocidad angular de la bobina.

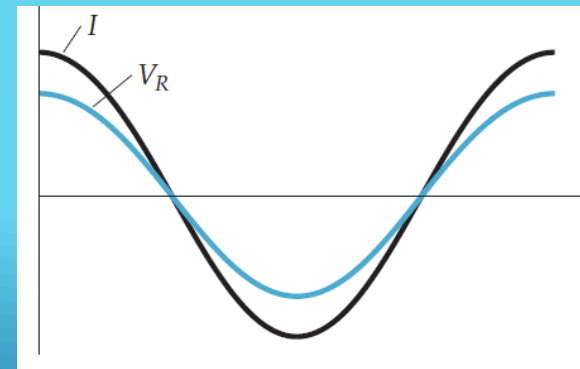
En la fig. de la derecha se muestra el circuito con una resistencia de consumidor donde

$$V_R = V_{R \text{ peak}} \cos \omega t$$

Al aplicar la ley de Ohm:

$$V_R = IR$$

$$V_{R \text{ peak}} \cos \omega t = IR$$



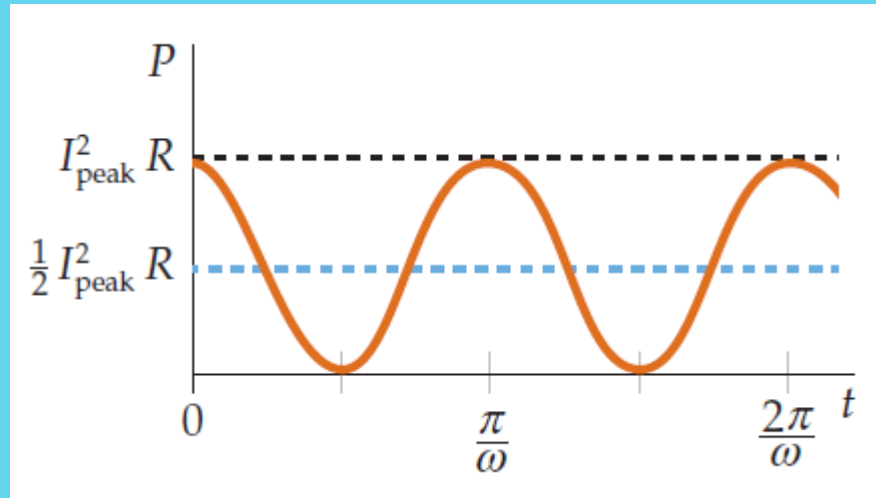
Corriente alterna con una resistencia

La potencia suministrada varía con el tiempo:

$$P = I^2 R = (I_{\text{peak}} \cos \omega t)^2 R = I_{\text{peak}}^2 R \cos^2 \omega t$$

$$P_{\text{av}} = (I^2 R)_{\text{av}} = I_{\text{peak}}^2 R (\cos^2 \omega t)_{\text{av}}$$

$$P_{\text{av}} = (I^2 R)_{\text{av}} = \frac{1}{2} I_{\text{peak}}^2 R$$



Valores de la raíz media cuadrática:

$$I_{\text{rms}} = \sqrt{(I^2)_{\text{av}}}$$

Para una corriente sinusoidal: $(I^2)_{\text{av}} = [(I_{\text{peak}} \cos \omega t)^2]_{\text{av}} = \frac{1}{2} I_{\text{peak}}^2$

Al sustituir este valor se obtiene:

$$I_{\text{rms}} = \frac{1}{\sqrt{2}} I_{\text{peak}} \approx 0.707 I_{\text{peak}}$$

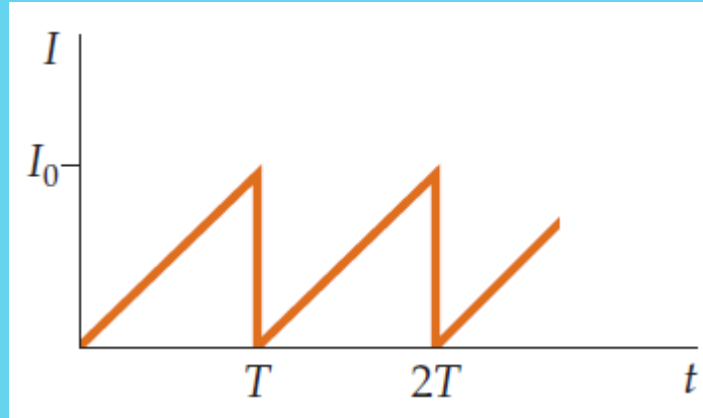
Onda diente de sierra

Encontrar la corriente promedio y la corriente rms de una onda diente de sierra de la figura.

$$I_{\text{av}} = \frac{1}{T} \int_0^T I dt = \frac{1}{T} \int_0^T \frac{I_0}{T} t dt = \frac{I_0}{T^2} \frac{T^2}{2} = \boxed{\frac{1}{2} I_0}$$

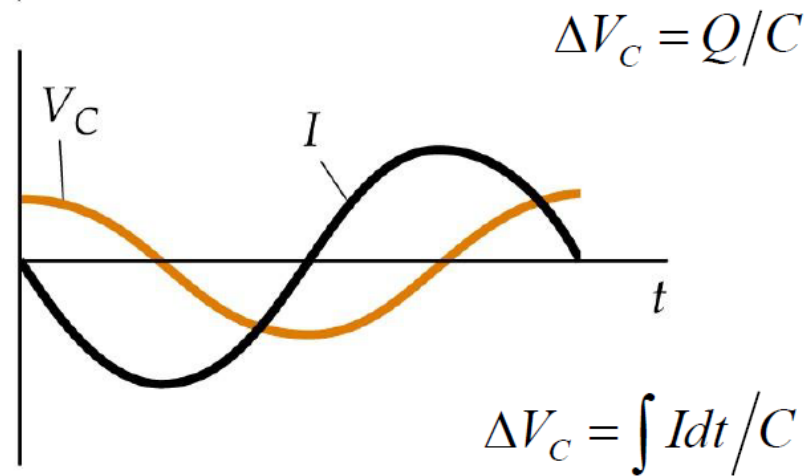
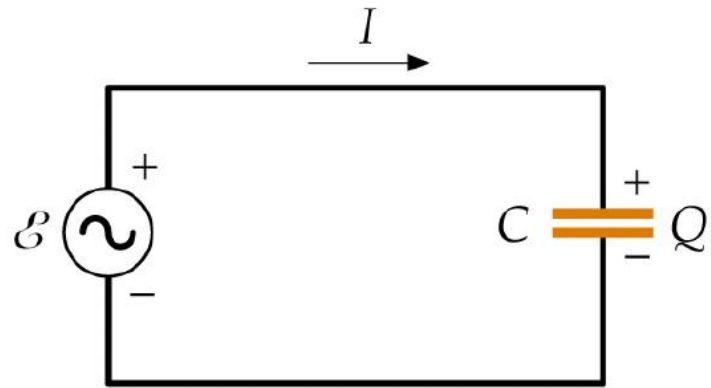
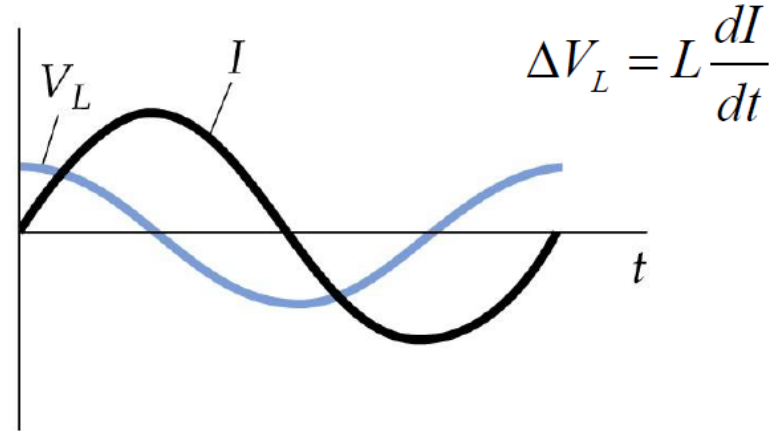
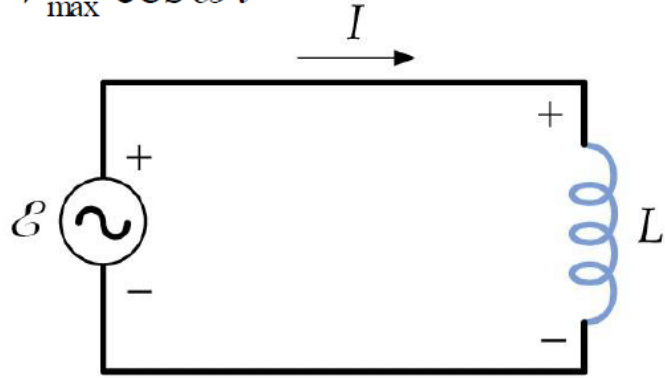
$$(I^2)_{\text{av}} = \frac{1}{T} \int_0^T I^2 dt = \frac{1}{T} \left(\frac{I_0}{T} \right)^2 \int_0^T t^2 dt = \frac{I_0^2}{T^3} \frac{T^3}{3} = \frac{1}{3} I_0^2$$

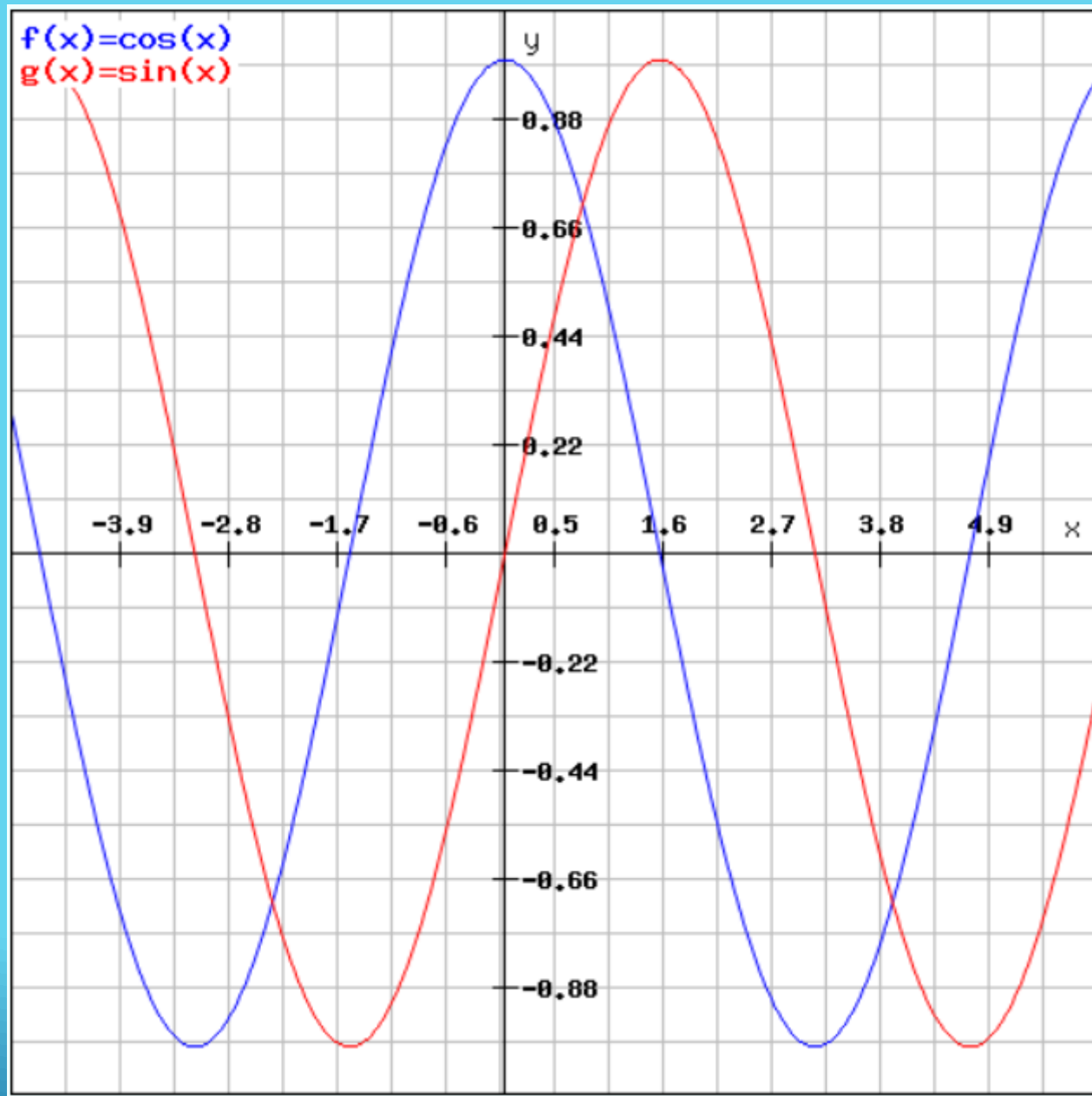
$$I_{\text{rms}} = \sqrt{(I^2)_{\text{av}}} = \boxed{\frac{I_0}{\sqrt{3}}}$$



Generador de corriente alterna con una bobina y con un condensador

$$\mathcal{E} = V_{\max} \cos \omega t$$





Reactancia inductiva

$$\mathcal{E} = V_{\max} \cos \omega t$$

$$I = I_{\max} \cos(\omega t - \delta)$$

$$\Delta V_L = L \frac{dI}{dt}$$

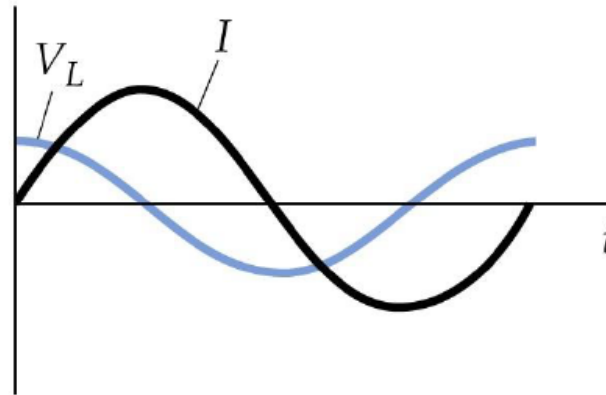
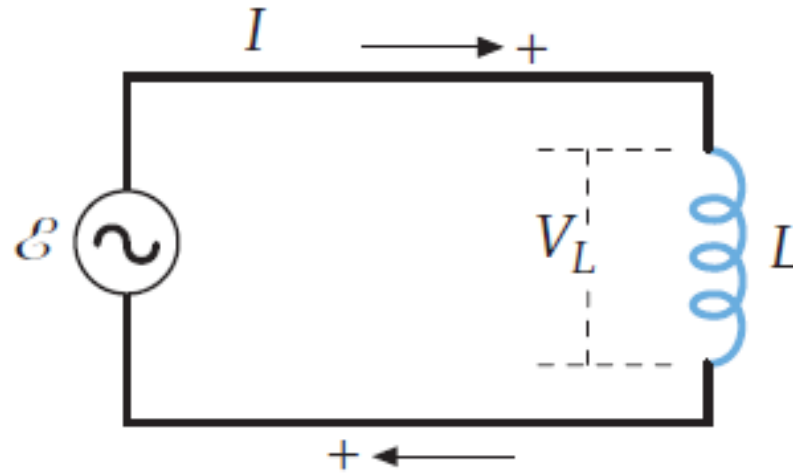
$$\Delta V_L = -LI_{\max} \omega \sin(\omega t - \delta)$$

$$\cos(\omega t + \pi/2) = -\sin(\omega t)$$

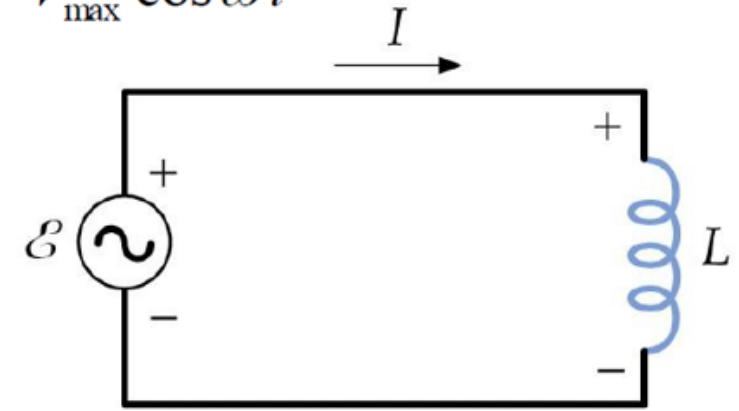
$$\Delta V_L = I_{\max} X_L \cos(\omega t - \delta + \pi/2)$$

$$X_L = L\omega$$

$$\Delta V_{L,\max} = I_{\max} X_L$$



$$\mathcal{E} = V_{\max} \cos \omega t$$



Reactancia capacitiva

$$\mathcal{E} = V_{\max} \cos \omega t$$

$$I = I_{\max} \cos(\omega t - \delta)$$

$$\Delta V_C = \int I dt / C$$

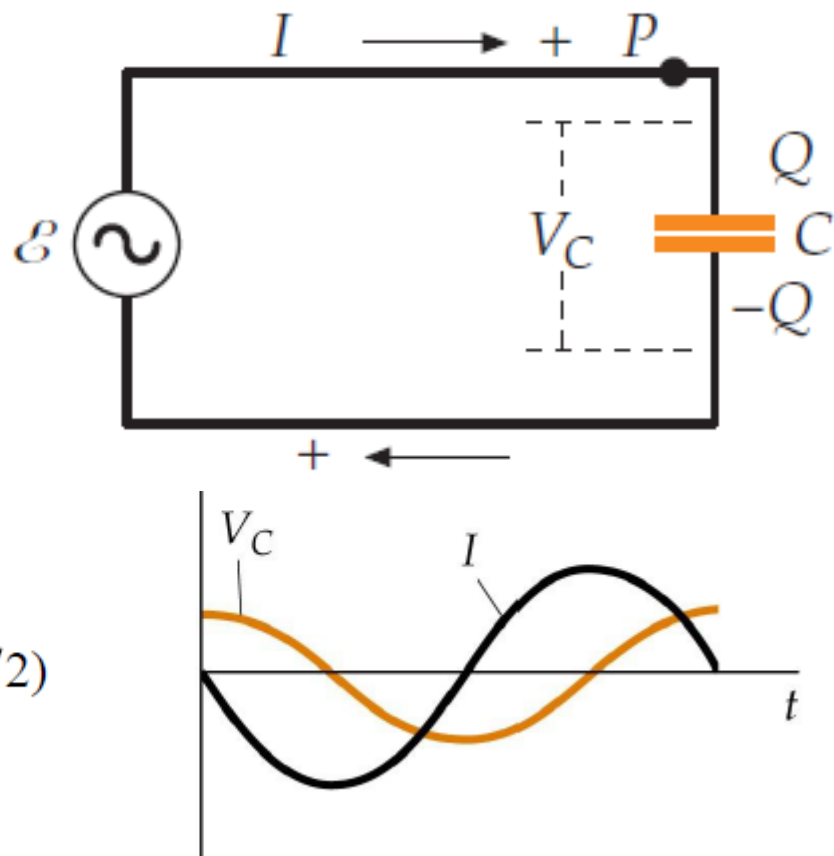
$$\Delta V_C = \frac{I_{\max}}{\omega C} \sin(\omega t - \delta)$$

$$\cos(\omega t - \pi/2) = \sin(\omega t)$$

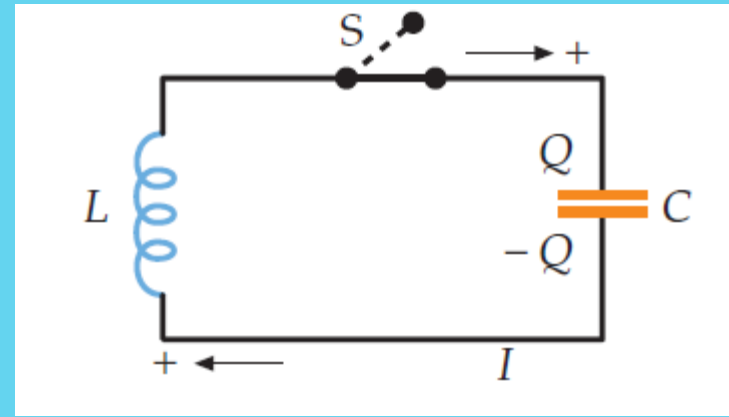
$$\Delta V_C = I_{\max} X_C \cos(\omega t - \delta - \pi/2)$$

$$X_C = 1/(\omega C)$$

$$\Delta V_{C,\max} = I_{\max} X_C$$



Circuitos RLC sin generador



$$I = + \frac{dQ}{dt}$$

$$L \frac{dI}{dt} + \frac{Q}{C} = 0$$

$$L \frac{d^2Q}{dt^2} + \frac{1}{C}Q = 0$$

$$m \frac{d^2x}{dt^2} + kx = 0$$

$$\frac{d^2Q}{dt^2} = -\frac{1}{LC}Q$$

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

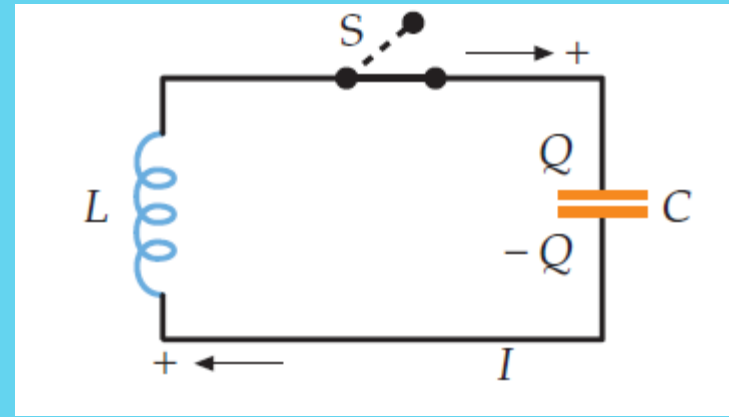
$$x = A \cos(\omega t - \delta)$$

$$Q = A \cos(\omega t - \delta)$$

$$\omega = \frac{1}{\sqrt{LC}}$$

Circuitos RLC sin generador

$$I = \frac{dQ}{dt} = -\omega A \sin(\omega t - \delta)$$



De las condiciones iniciales $Q = Q_{\text{pico}}$ e $I = 0$ en $t = 0$, la constante de fase es cero y $A = Q_{\text{pico}}$. Entonces:

$$Q = Q_{\text{peak}} \cos \omega t$$

$$I = -\omega Q_{\text{peak}} \sin \omega t = -I_{\text{peak}} \sin \omega t$$

La energía eléctrica almacenada en el capacitor:

$$U_e = \frac{1}{2} Q V_c = \frac{1}{2} \frac{Q^2}{C}$$

$$U_e = \frac{1}{2} \frac{Q_{\text{peak}}^2}{C} \cos^2 \omega t$$

La energía magnética almacenada en la inductancia:

$$U_m = \frac{1}{2} L I^2$$

Circuitos RLC sin generador

Sustituyendo $I = -\omega Q_{\text{pico}}$:

$$U_m = \frac{1}{2} L \omega^2 Q_{\text{peak}}^2 \sin^2 \omega t = \frac{1}{2} \frac{Q_{\text{peak}}^2}{C} \sin^2 \omega t$$

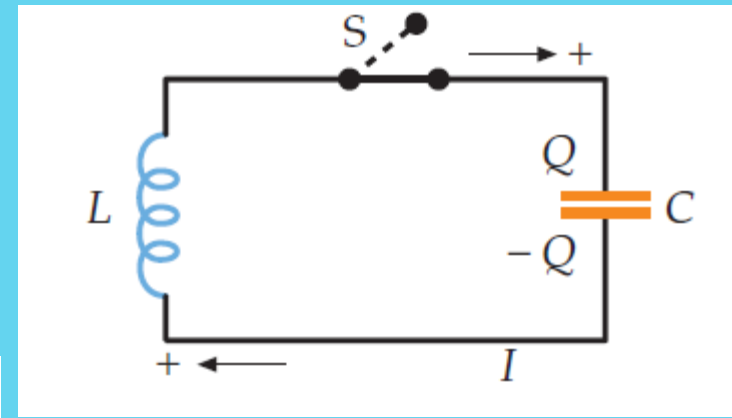
$$U_{\text{total}} = U_e + U_m = \frac{1}{2} \frac{Q_{\text{peak}}^2}{C} \cos^2 \omega t + \frac{1}{2} \frac{Q_{\text{peak}}^2}{C} \sin^2 \omega t = \frac{1}{2} \frac{Q_{\text{peak}}^2}{C}$$

En un circuito RLC tendremos:

$$L \frac{dI}{dt} + IR + \frac{Q}{C} = 0$$

$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{1}{C} Q = 0$$

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = 0$$

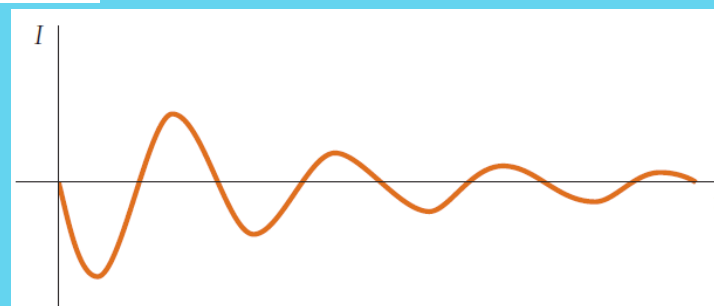
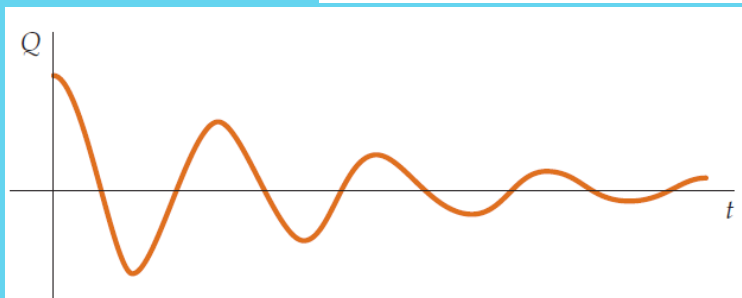
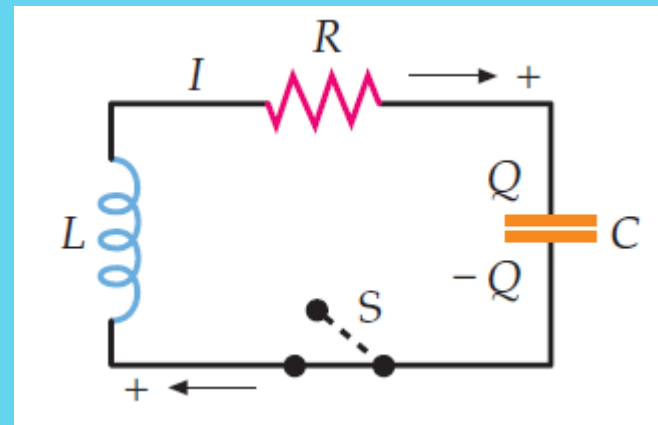


Circuitos RLC sin generador

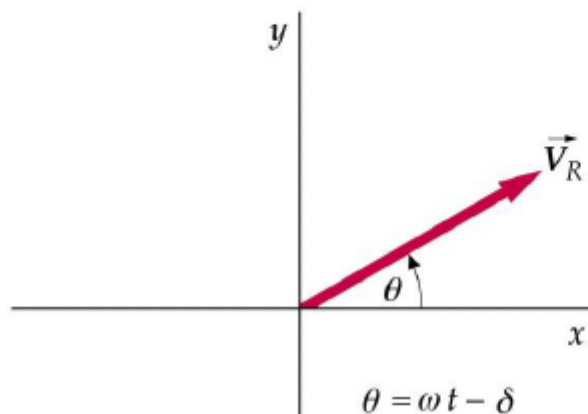
Si la resistencia es pequena se tiene la frecuencia propia del oscilador

$$\omega_0 = 1/\sqrt{LC}$$

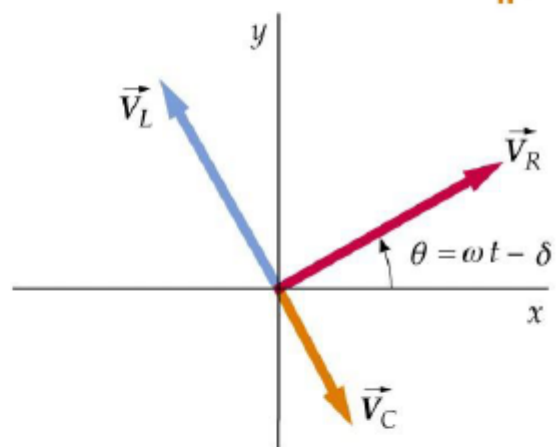
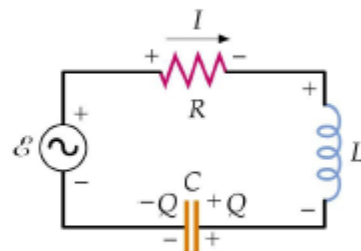
$$LI\frac{dI}{dt} + I^2R + I\frac{Q}{C} = 0$$



Representación fasorial

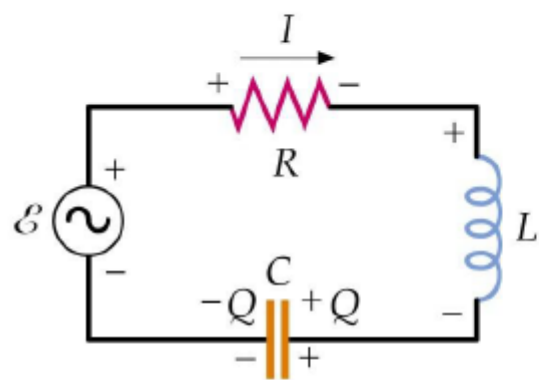


$$\varepsilon = V_{\max} \cos \omega t$$



Representación de las tensiones en una resistencia, bobina y condensador, cada vector gira en sentido anti-horario con frecuencia ω

Circuito RLC con generador de ac



$$\Delta V_R = RI$$

$$\Delta V_C = Q/C$$

$$\Delta V_L = L \frac{dI}{dt}$$

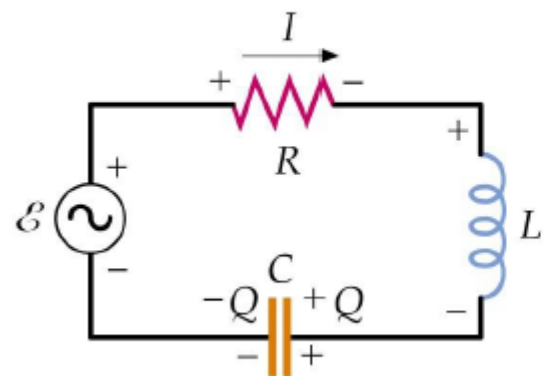
$$L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = V_{\max} \cos \omega t$$

Oscilador armónico amortiguado y forzado

$$Q = Q_{\max} \sin(\omega t - \delta)$$

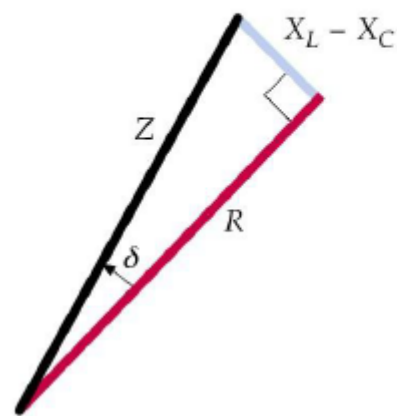
$$I = I_{\max} \cos(\omega t - \delta)$$

Reactancias en Circuito RLC



$$I = I_{\max} \cos(\omega t - \delta)$$

$$\tan \delta = \frac{\omega L - 1/(\omega C)}{R} = \frac{X_L - X_C}{R}$$



$$X_L = \omega L$$

Reactancia inductiva y
capacitiva (Unidad: Ω)

$$X_C = 1/(\omega C)$$

$$X = X_L - X_C$$

Reactancia total

$$Z = \sqrt{R^2 + X^2}$$

Impedancia

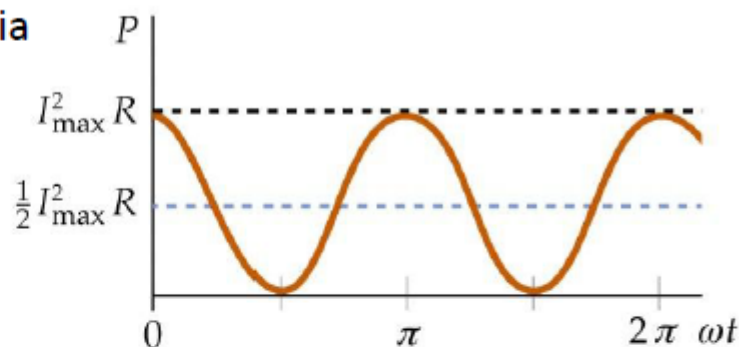
$$I_{\max} = \frac{V_{\max}}{Z}$$

Potencia suministrada y factor de potencia

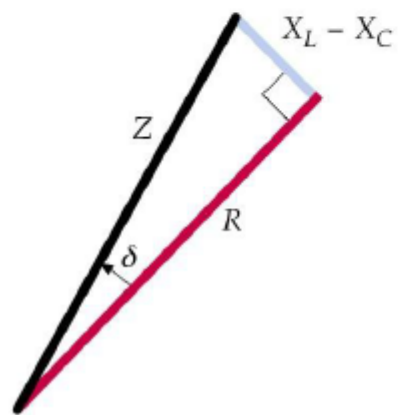
Potencia instantánea disipada en la resistencia

$$P = RI^2 = RI_{\max}^2 \cos^2(\omega t - \delta)$$

La potencia media es $P_m = \frac{1}{2} RI_{\max}^2$



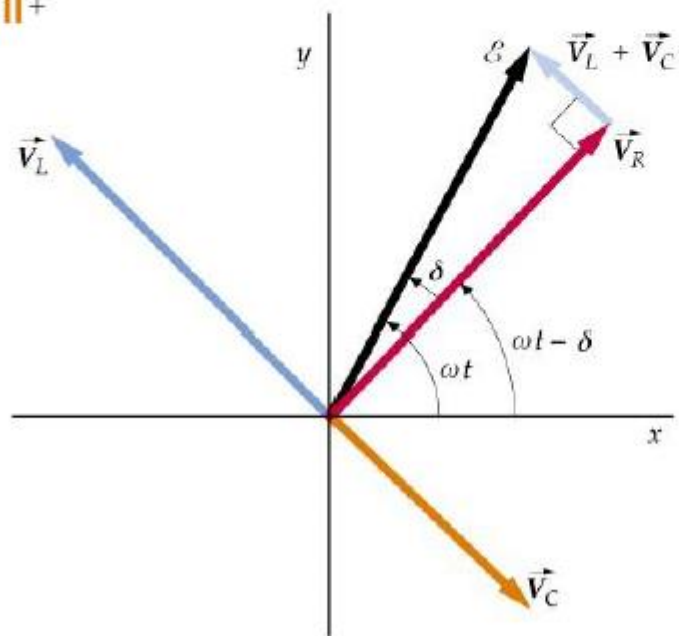
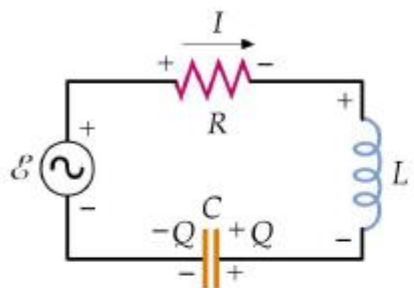
$$I_{\max} = \frac{V_{\max}}{Z} \quad P_m = \frac{1}{2} \frac{R}{Z} I_{\max} V_{\max}$$



$$\cos \delta = \frac{R}{Z} \Rightarrow P_m = \frac{1}{2} I_{\max} V_{\max} \cos \delta$$

$\cos \delta$ es el factor de potencia

Representación fasorial



$$\mathcal{E} = V_{\max} \cos \omega t \quad \text{generador}$$

$$I = I_{\max} \cos(\omega t - \delta)$$

$$V_R = RI$$

$$V_L = L \frac{dI}{dt}$$

$$V_C = \frac{Q}{C}$$

Resonancia

$$X_L = X_C \quad Z = \sqrt{R^2 + X^2} = R \quad \text{En resonancia la impedancia es mínima}$$

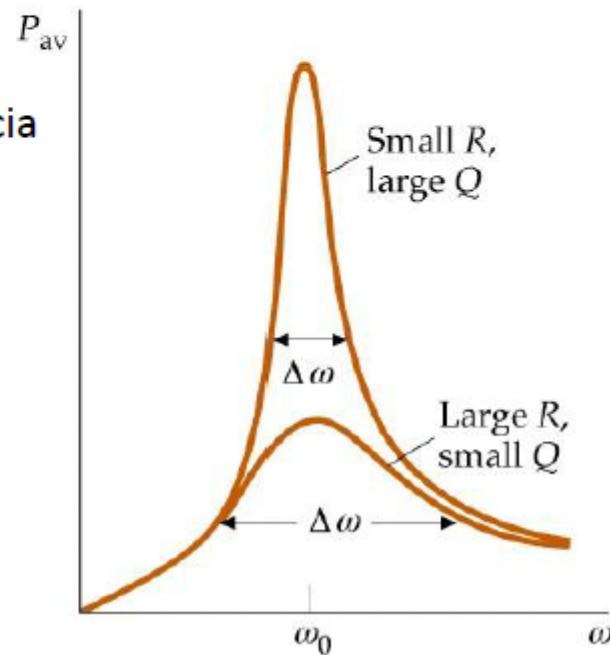
$$\left. \begin{array}{l} X_L = \omega L \\ X_C = 1/(\omega C) \end{array} \right\} \Rightarrow \omega = \frac{1}{\sqrt{LC}} \quad \text{En resonancia la frecuencia del generador coincide con la frecuencia natural del circuito.}$$

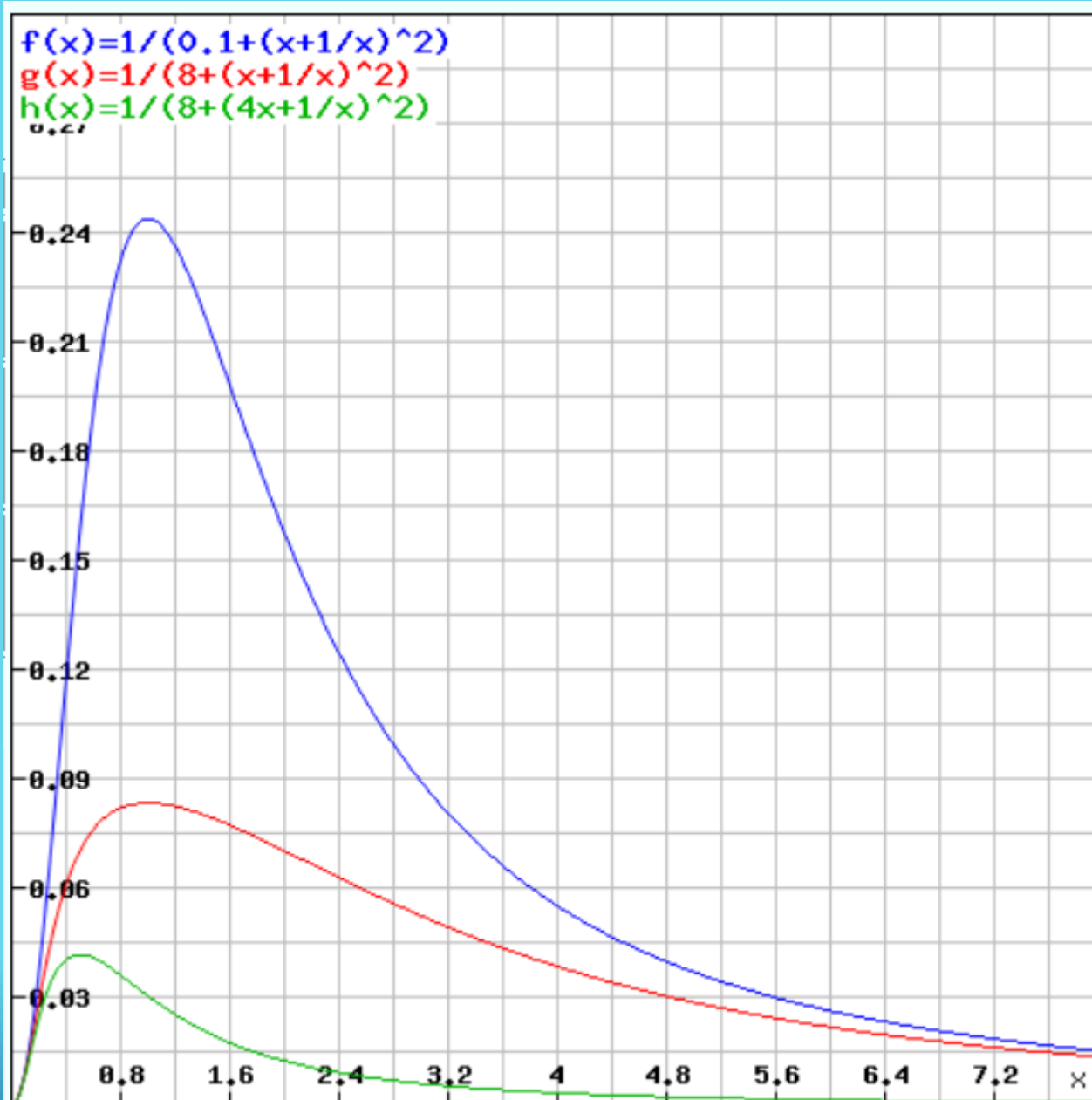
$$\tan \delta = \frac{X_L - X_C}{R} = 0 \Rightarrow \delta = 0 \quad \text{En resonancia el factor de potencia (cos } \delta) \text{ es } = 1$$

$$P_m = \frac{1}{2} \frac{R}{Z} I_{\max} V_{\max} \quad I_{\max} = \frac{V_{\max}}{Z}$$

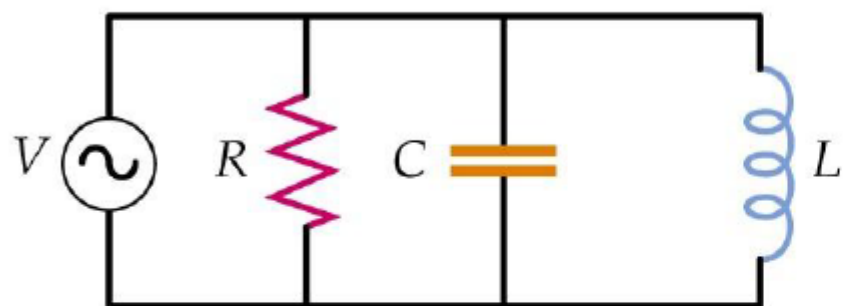
$$P_m = \frac{1}{2} \frac{R V_{\max}^2}{Z^2} = \frac{1}{2} \left\{ \frac{R V_{\max}^2}{R^2 + [\omega L - 1/(\omega C)]^2} \right\}$$

$$Q = \frac{\omega_0}{\Delta \omega} = \frac{\omega_0 L}{R} \quad \text{Factor de calidad}$$





Circuito R, L y C en paralelo

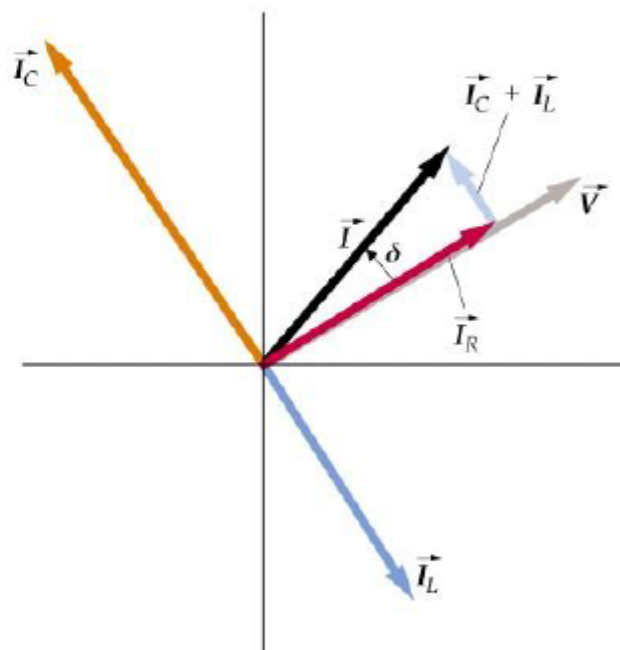


La diferencia de potencial instantánea es la misma en los tres elementos

$$I = \sqrt{I_R^2 + (I_C - I_L)^2}$$

$$I = \sqrt{\left(\frac{V}{R}\right)^2 + \left(\frac{V}{X_C} - \frac{V}{X_L}\right)^2}$$

$$I = \frac{V}{Z}$$



$$\frac{1}{Z} = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_C} - \frac{1}{X_L}\right)^2}$$

$$I = \sqrt{I_R^2 + (I_L - I_C)^2} = \sqrt{\left(\frac{V_{\text{máx}}}{R}\right)^2 + \left(\frac{V_{\text{máx}}}{X_L} - \frac{V_{\text{máx}}}{X_C}\right)^2} = \frac{V_{\text{máx}}}{Z}$$

Valores eficaces

$$\varepsilon = V_{\max} \cos \omega t$$

(tensión y corriente instantáneas)

$$I = I_{\max} \cos(\omega t - \delta)$$

Definición del valor eficaz de la corriente: $I_{ef} = \sqrt{\langle I^2(t) \rangle}$ $\langle I^2(t) \rangle = \frac{1}{T} \int_t^{t+T} I^2(t) dt$ $T = \frac{1}{f} = \frac{2\pi}{\omega}$

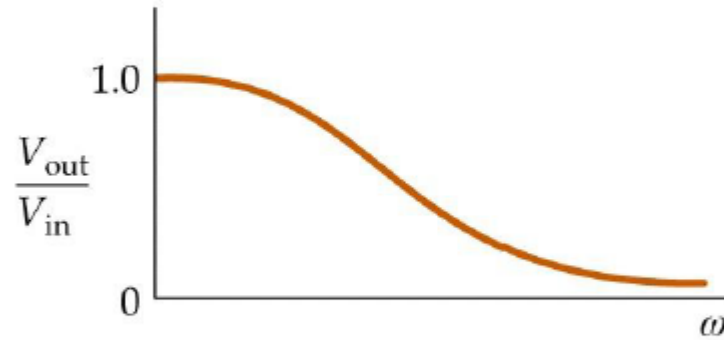
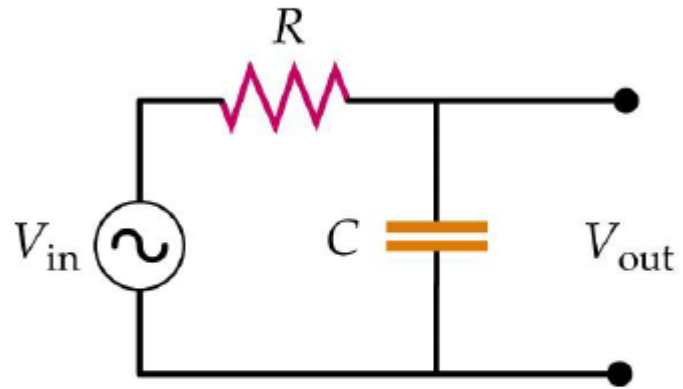
$$\langle I^2(t) \rangle = \frac{1}{T} \int_t^{t+T} I_{\max}^2 \cos^2(\omega t - \delta) dt = \frac{I_{\max}^2}{2} \Rightarrow I_{ef} = \frac{I_{\max}}{\sqrt{2}} = 0.707 I_{\max}$$

$$\varepsilon = V_{\max} \cos \omega t \Rightarrow \varepsilon_{ef} = \frac{V_{\max}}{\sqrt{2}} \text{ valor eficaz de la tensión}$$

Amperímetros y voltímetros en corriente alterna están diseñado para medir valores eficaces. La potencia media suministrada por un generador es:

$$P_m = \langle P \rangle = R \langle I^2 \rangle = R I_{\max}^2 \langle \cos^2(\omega t - \delta) \rangle \Rightarrow P_m = \frac{1}{2} R I_{\max}^2 = R I_{ef}^2 = \varepsilon_{ef} I_{ef}$$

Ejemplo: filtro “pasa-baja”



$$\frac{V_{out}}{V_{in}} = \frac{1}{\sqrt{1 + (RC\omega)^2}}$$

$$V_{sal\ ef} = I_{ef} X_C$$

$$I_{ef} = \frac{V_{ap\ ef}}{Z}$$

$$Z = \sqrt{R^2 + X_C^2}$$

$$V_{sal\ ef} = I_{ef} X_C = \frac{V_{ap\ ef} X_C}{\sqrt{R^2 + X_C^2}} = \frac{V_{ap\ ef}}{\sqrt{1 + \frac{R^2}{X_C^2}}} = \boxed{\frac{V_{ap\ ef}}{\sqrt{1 + (\omega RC)^2}}}$$

Problemas:

P. A. Tipler y G. Mosca, Physics for Scientist and Engineers with Modern Physics, Sixth Edition, 2008,

Math tutorial, Answer to odd numbers End of Chapter Problems.

Tipler

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